

AJAY KUMAR PONDUGALA

Backend reliability and service infrastructure - recruiting proof packet

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FIT IN 30 SECONDS

Software Engineer II at Cisco, based in Bengaluru. Full-time since August 2024, with a separate Jan-Jun 2024 internship. Target: backend/platform/distributed-systems roles in India. Core evidence: Java/Go services, messaging, PostgreSQL, Kubernetes, regression validation and observability.

TWO ACCEPTED RELIABILITY FIXES

NATS Python - safe credential serialization | [PR #1043](#)

Problem: a signed CONNECT path omitted an explicit password. Contribution: preserve both password and signed credentials; regression coverage for authentication serialization and refreshed credentials. Maintainer reviewed and merged 2026-10-01 21:36:34 UTC. This is client serialization evidence, not a claim of production authentication deployment.

Prometheus - alert continuity after restart | [PR #19882](#)

Problem: firing alerts with a short for duration could wait again after restart. Contribution: restore the previous active timestamp while preserving pending-alert/grace-period semantics. Firing resumes on the next scheduled evaluation. Maintainer reviewed and merged 2026-10-05 10:20:20 UTC.

FLAGSHIP: DURABLE ASYNCHRONOUS BACKEND

[Code and clean-clone instructions](#) | [Case study](#)

Implemented and tested: separate Go API, outbox dispatcher and workers; PostgreSQL durable acceptance/idempotency and results; RabbitMQ confirmed publication; commit before acknowledgement; bounded retries/dead letters. Recovery regressions cover worker death before commit and after commit-before-ack, dispatcher death after publish, broker restart, duplicates and database errors.

Measured: 1/2/4/8 independent workers, three seeded 1,000-job trials per count, plus warm-up. 262.73-323.21 jobs/s; observed p95 95.43-151.08 ms (submission through first committed-result observation, including polling). Across four isolated schemas: 12,400 accepted = completed = durable results; pending outbox zero. Broker restart recovery 3.402 s includes Compose startup and health wait.

Limits: synthetic single-host Docker lab, not production or distributed-scale proof. The earlier 10,000-job result is in-memory functional evidence. [Corrected raw results and provenance](#) describe polling/resource overhead and source attribution.

SECOND CASE STUDY: DETERMINISTIC RECOVERY

[Transformer recovery lab](#) - CPU checkpoint/restart tests and reproducible public-data experiments. GPU, distributed-scale and production evidence are not claimed.

REVIEW THE EVIDENCE

Start with the two merged PRs, then the flagship's commands, raw receipts and [90-second recovery test replay](#). [Canonical portfolio](#) has the case studies. The captioned video is actual agent-executed output, not Ajay's presentation or evidence of his personal execution.